

PART II - INDUSTRIAL PLC PROJECTS

Project 3 - Water Tank Pump Control System

Version 2 - Industrial PLC Upgrade

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite

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1. Project Overview

Project 3 Version 2 develops the original water-tank control concept into a more structured PLC pump-control application. The final program includes Auto and Manual operating modes, operating permissives, System Ready status, automatic level-based pump requests, manual pump control, delayed pump startup, low- and high-level indication, dry-run monitoring, master fault handling, and operator alarm indication.

I developed the control program using an Allen-Bradley MicroLogix 1100 PLC in RSLogix Micro Starter Lite. Rather than driving the pump directly from one level switch, the Version 2 design separates operating conditions, mode-specific requests, the common Pump Run Request, startup timing, and the physical Pump_Motor O:2/0 output.

In Auto mode, the program uses Low_Level_Switch I:1/7 and High_Level_Switch I:1/8 to determine when an automatic pump request can be established. Auto_Run_Request B3:0/0 is latched and is removed when the high-level condition is reached, the implemented Stop condition occurs, a master fault becomes active, or Auto mode is removed.

Manual operation is implemented separately. Manual_Run_Request B3:0/1 is generated from Manual mode, Start_PB I:1/0, and System Ready. The Auto and Manual requests are then combined into Pump_Run_Request B3:0/2. A three-second Delay_Timer T4:0 is applied before Pump_Motor O:2/0 energizes, and Running_Lamp O:2/1 follows the commanded pump state.

The program also provides direct Low Level and High Level indication through O:2/4 and O:2/5. Dry-run monitoring uses Dry_Run_Timer T4:1. While Pump_Motor is ON and Low_Level_Switch is false, the timer accumulates toward a ten-second preset. When the timer reaches Done, Dry_Run_Fault B3:1/0 is latched and participates in the master Fault_Active B3:0/4 logic.

This project was developed and tested as a PLC portfolio/training implementation. The E-stop and overload signals represented in the program are standard PLC monitoring conditions and are not presented as a complete safety-rated machine protection system.

2. Engineering Objective

The objective of Project 3 Version 2 was to upgrade the original tank-level control project into a structured pump-control system using operating modes, permissives, level-based automatic control, fault monitoring, and clear operating-status signals.

- Provide separate Auto and Manual operating modes.
- Detect simultaneous Auto/Manual mode selection.
- Monitor E-stop and overload healthy conditions.
- Combine major abnormal conditions into a latched master fault.
- Provide a centralized Start Permissive and System Ready condition.
- Automatically request pump operation from the implemented level-switch conditions.
- Stop the automatic pump request when the high-level condition is reached.
- Provide a separate Manual pump request using the operator Start command.
- Combine Auto and Manual requests into one common Pump Run Request.
- Apply a three-second delay before the physical pump output energizes.
- Provide Running, Low Level, High Level, Fault, and Alarm indications.
- Detect a possible dry-run condition while the pump is operating.
- Latch a Dry Run Fault after the configured ten-second timeout.

3. Functional Requirements

- The PLC shall monitor Auto_Mode I:1/3 and Manual_Mode I:1/4.
- The PLC shall energize Mode_Conflict B3:0/5 when Auto and Manual modes are selected simultaneously.
- Fault_Active B3:0/4 shall latch from the implemented Mode Conflict, unhealthy E-stop, unhealthy overload, or Dry Run Fault condition.
- Fault_Active shall be reset only when Reset_PB I:1/2 is active, EStop_OK and Overload_OK are healthy, and Mode_Conflict is false.
- Start_Permissive B3:0/7 shall become true when Fault_Active is false, EStop_OK and Overload_OK are true, and Mode_Conflict is false.
- System_Ready B3:0/3 shall require Start_Permissive and an Auto or Manual mode selection.
- In Auto mode, Auto_Run_Request B3:0/0 shall latch when System Ready is true, Low_Level_Switch I:1/7 is true, and High_Level_Switch I:1/8 is false.
- Auto_Run_Request shall unlatch from the implemented High Level, Stop, Fault Active, or Auto-mode-removal conditions.
- In Manual mode, Manual_Run_Request B3:0/1 shall energize while Start_PB and System_Ready are true.
- Pump_Run_Request B3:0/2 shall become true from either Auto or Manual request while System_Ready remains true.
- Delay_Timer T4:0 shall use time base 1.0 s and preset 3.
- Pump_Motor O:2/0 shall energize when T4:0/DN and Pump_Run_Request are true.
- Running_Lamp O:2/1 shall follow Pump_Motor.
- Low_Level_Lamp O:2/4 and High_Level_Lamp O:2/5 shall follow their corresponding level switches.
- Dry_Run_Timer T4:1 shall run while Pump_Motor is ON and Low_Level_Switch is false, using time base 1.0 s and preset 10.
- Dry_Run_Fault B3:1/0 shall latch when T4:1/DN becomes true.
- Fault_Lamp O:2/2 and Alarm_Horn O:2/3 shall energize while Fault_Active is true.

The supplied final ladder does not show an OTU instruction for Dry_Run_Fault B3:1/0. The report therefore preserves that implementation exactly and does not claim a Dry Run Fault reset function that is not visible in the final ladder.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes pump-control and monitoring logic
Programming Software	RSLogix Micro Starter Lite	Ladder development and PLC testing
Start Push Button	I:1/0	Manual-mode Start command
Stop Input	I:1/1	Participates in automatic request reset logic
Reset Push Button	I:1/2	Master-fault reset command
Auto Mode Selector	I:1/3	Selects automatic tank-level control
Manual Mode Selector	I:1/4	Selects manual pump operation
E-stop Healthy Input	I:1/5	E-stop circuit healthy status
Overload Healthy Input	I:1/6	Motor overload healthy status
Low Level Switch	I:1/7	Low-level process status
High Level Switch	I:1/8	High-level process status
Pump Motor	O:2/0	Pump motor command
Running Lamp	O:2/1	Commanded pump indication
Fault Lamp	O:2/2	Master-fault indication
Alarm Horn	O:2/3	Audible master-fault indication

Component	Identification	Function
Low Level Lamp	O:2/4	Low-level indication
High Level Lamp	O:2/5	High-level indication
Delay Timer	T4:0	Three-second startup delay
Dry Run Timer	T4:1	Ten-second dry-run monitoring timer

5. I/O Assignment

Physical Inputs

Device	PLC Address	Type	Description
Start_PB	I:1/0	Digital Input	Manual-mode Start command
Stop_PB	I:1/1	Digital Input	Stop input in Auto request reset logic
Reset_PB	I:1/2	Digital Input	Fault-reset command
Auto_Mode	I:1/3	Digital Input	Auto mode selector
Manual_Mode	I:1/4	Digital Input	Manual mode selector
EStop_OK	I:1/5	Digital Input	E-stop healthy status
Overload_OK	I:1/6	Digital Input	Overload healthy status
Low_Level_Switch	I:1/7	Digital Input	Low-level status
High_Level_Switch	I:1/8	Digital Input	High-level status

Physical Outputs

Device	PLC Address	Type	Description
Pump_Motor	O:2/0	Digital Output	Pump motor command
Running_Lamp	O:2/1	Digital Output	Pump running command indication
Fault_Lamp	O:2/2	Digital Output	Master-fault indication
Alarm_Horn	O:2/3	Digital Output	Audible fault indication
Low_Level_Lamp	O:2/4	Digital Output	Low-level indication
High_Level_Lamp	O:2/5	Digital Output	High-level indication

Internal PLC Data

Function	Address	Type	Purpose
Auto_Run_Request	B3:0/0	Internal Bit	Latched automatic pump request
Manual_Run_Request	B3:0/1	Internal Bit	Non-latched Manual pump request
Pump_Run_Request	B3:0/2	Internal Bit	Common pump operating request
System_Ready	B3:0/3	Internal Bit	Operating readiness status
Fault_Active	B3:0/4	Latched Internal Bit	Master fault condition
Mode_Conflict	B3:0/5	Internal Bit	Simultaneous mode selection
Start_Permissive	B3:0/7	Internal Bit	Master pump-start permission
Dry_Run_Fault	B3:1/0	Latched Internal Bit	Dry-run diagnostic fault
Delay_Timer	T4:0	TON	Three-second startup delay
Dry_Run_Timer	T4:1	TON	Ten-second dry-run timer

6. Control Strategy

I organized the Version 2 tank-control program around five primary functions: operating validity, mode-specific pump requests, common pump control, level indication, and dry-run/fault monitoring.

The program first evaluates the operating mode and abnormal conditions. Simultaneous Auto and Manual selection produces Mode Conflict. The master Fault Active latch combines Mode Conflict, unhealthy E-stop status, unhealthy overload status, and Dry Run Fault.

Start Permissive provides a centralized operating-permission signal. System Ready requires Start Permissive together with an active Auto or Manual mode selection.

Automatic operation is level-based. In Auto mode, System Ready, Low Level Switch, and the not-High-Level condition latch Auto Run Request. A dedicated unlatch rung removes the request when High Level Switch becomes true, the implemented Stop condition occurs, Fault Active becomes true, or Auto mode is removed.

Manual operation uses an OTE rather than a latch and therefore exists only while Manual mode, Start_PB, and System Ready remain true. Auto and Manual requests are combined into Pump Run Request so both modes can share the same startup timer and physical output logic.

Delay_Timer T4:0 provides a three-second delay before pump energization. Separate Low Level and High Level lamps indicate the corresponding switch states.

Dry-run monitoring operates while Pump Motor is ON and Low Level Switch is false. If that condition remains for ten seconds, T4:1/DN latches Dry Run Fault. Dry Run Fault then participates in the master Fault Active logic, which creates the visual and audible fault response.

7. Engineering Design Decisions

Design Decision	Engineering Reason
Separate Auto and Manual operating modes	Supports automatic level-based control and operator-controlled Manual operation
Dedicated Mode Conflict bit	Explicitly identifies simultaneous mode selection as invalid
Master Fault Active latch	Consolidates implemented abnormal conditions into one persistent fault state
Central Start Permissive	Creates one clear operating-permission status
Separate System Ready bit	Distinguishes operating readiness from an actual pump request
Level-based Auto Run Request	Allows the tank process to request pumping from the implemented level conditions
Latched Auto Run Request	Maintains automatic pump demand after the initiating condition is detected
Dedicated Auto request unlatch rung	Defines conditions that terminate automatic pumping
Non-latched Manual Run Request	Keeps Manual operation dependent on continuing operator command
Common Pump Run Request	Allows both modes to share startup and output logic
Three-second Delay Timer	Prevents immediate pump energization
Dedicated Pump Motor output rung	Provides one clear physical output-control point
Low/High Level lamps	Provides direct process-state indication
Dry Run Timer	Detects prolonged pump operation without the expected Low Level Switch state
Latched Dry Run Fault	Preserves the detected abnormal event
Fault Lamp and Alarm Horn	Provides consistent visual and audible master-fault indication

One of the main Version 2 improvements is the separation between process demand and equipment output. The level switches do not directly control Pump_Motor O:2/0; they establish an internal Auto Run Request that passes through common request and timing logic before the physical output energizes.

The final ladder also includes a latched Dry_Run_Fault B3:1/0. Because the supplied final screenshots do not show an OTU for that bit, this report preserves the implemented behavior rather than silently adding a reset path.

8. Ladder Logic Description

The final Project 3 Version 2 program contains 18 functional rungs, Rung 0000 through Rung 0017, followed by END at Rung 0018. Each rung below is reconstructed from the final RSLogix screenshots without changing addresses, instructions, branches, timer settings, or implemented behavior.

8.1 Rung 0000 - Mode Conflict Detection

Ladder Logic

```
|-----[ XIC I:1/3 ]-----[ XIC I:1/4 ]------( OTE B3:0/5 )-----|
      Auto_Mode      Manual_Mode      Mode_Conflict
```

Rung Description:

Rung 0000 monitors the Auto and Manual mode selectors. When both inputs are true, Mode_Conflict B3:0/5 energizes. Because an OTE is used, the bit follows the current simultaneous-selection condition.

Engineering Purpose:

I included this rung to explicitly identify an invalid operating-mode combination and make that condition available to the master fault, reset, and permissive logic.

8.2 Rung 0001 - Master Fault Latch

Ladder Logic

```

+----[ XIC B3:0/5 ]-----+
|      Mode_Conflict      |
+----[ XIO I:1/5 ]-----+
|      EStop_OK          |
|-----+----[ XIO I:1/6 ]-----+----( OTL B3:0/4 )-----|
|      Overload_OK      |      Fault_Active
+----[ XIC B3:1/0 ]-----+
|      Dry_Run_Fault    |

```

Rung Description:

Four parallel conditions can latch Fault_Active B3:0/4: Mode Conflict, loss of EStop_OK, loss of Overload_OK, or Dry_Run_Fault. The OTL keeps the master fault stored after the initiating condition disappears.

Engineering Purpose:

I used one master fault bit to consolidate the implemented abnormal conditions into a common shutdown and alarm state.

8.3 Rung 0002 - Master Fault Reset

Ladder Logic

```
|-----[ XIC I:1/2 ]-----[ XIC I:1/5 ]-----[ XIC I:1/6 ]-----[ XIO B3:0/5 ]----- ( OTU B3:0/4 )-----|
      Reset_PB      EStop_OK      Overload_OK      Mode_Conflict      Fault_Active
```

Rung Description:

Reset can unlatch Fault_Active only when Reset_PB is true, EStop_OK and Overload_OK are healthy, and Mode_Conflict is false. The final screenshot shows an OTU for Fault_Active only.

Engineering Purpose:

I used conditional reset logic so the master fault cannot be cleared while the monitored E-stop, overload, or mode-conflict condition remains abnormal.

Implementation Note:

A separate OTU for Dry_Run_Fault B3:1/0 is not visible in the final ladder.

8.4 Rung 0003 - Start Permissive

Ladder Logic

```
|-----[ XIO B3:0/4 ]-----[ XIC I:1/5 ]-----[ XIC I:1/6 ]-----[ XIO B3:0/5 ]----- ( OTE B3:0/7 )-----|
      Fault_Active      EStop_OK      Overload_OK      Mode_Conflict      Start_Permissive
```

Rung Description:

Start_Permissive becomes true only when Fault_Active is false, EStop_OK and Overload_OK are true, and Mode_Conflict is false.

Engineering Purpose:

I centralized the primary operating conditions into one permissive bit rather than repeating them throughout the pump-control sequence.

8.5 Rung 0004 - System Ready

Ladder Logic

```

      +----[ XIC I:1/3 ]-----+
      |      Auto_Mode      |
|-----[ XIC B3:0/7 ]-----+-----+ ( OTE B3:0/3 )-----|
      Start_Permissive      |      System_Ready
      +----[ XIC I:1/4 ]-----+
      |      Manual_Mode     |

```

Rung Description:

Start Permissive is in series with a parallel Auto/Manual branch. System_Ready becomes true when operation is permitted and at least one mode is selected.

Engineering Purpose:

System Ready provides a separate status indicating that the program is permitted to operate and an operating mode has been selected.

8.6 Rung 0005 - Automatic Pump Run Request

Ladder Logic

```
|-----[ XIC I:1/3 ]-----[ XIC B3:0/3 ]-----[ XIC I:1/7 ]-----[ XIO I:1/8 ]----- ( OTL B3:0/0 )-----|
      Auto_Mode      System_Ready      Low_Level_Switch      High_Level_Switch      Auto_Run_Request
```

Rung Description:

In Auto mode, the request can latch when System Ready is true, Low Level Switch is true, and High Level Switch is false. Because OTL is used, the request remains stored until its unlatch rung executes.

Engineering Purpose:

I used a latched internal request so process demand is separated from direct physical pump control.

8.7 Rung 0006 - Automatic Pump Run Request Reset

Ladder Logic

```

+----[ XIC I:1/8 ]-----+
|   High_Level_Switch   |
+----[ XIO I:1/1 ]-----+
|   Stop_PB             |
|-----+----[ XIC B3:0/4 ]-----+----( OTU B3:0/0 )----|
|   Fault_Active        |   Auto_Run_Request
+----[ XIO I:1/3 ]-----+
|   Auto_Mode

```

Rung Description:

Any of four parallel conditions unlatches Auto_Run_Request: High Level Switch true, the implemented Stop condition true through XIO I:1/1, Fault Active true, or Auto mode removed.

Engineering Purpose:

This dedicated unlatch rung clearly defines the implemented conditions that terminate the maintained automatic pump request.

8.8 Rung 0007 - Manual Pump Run Request

Ladder Logic

```

|----[ XIC I:1/4 ]----[ XIC I:1/0 ]----[ XIC B3:0/3 ]----- ( OTE B3:0/1 )----|
|   Manual_Mode      |   Start_PB      |   System_Ready      |   Manual_Run_Request

```

Rung Description:

Manual_Run_Request becomes true while Manual mode, Start_PB, and System Ready are all true. Because an OTE is used, the request disappears when any enabling condition becomes false.

Engineering Purpose:

I kept Manual operation non-latched so it depends on a continuing operator command.

8.9 Rung 0008 - Common Pump Run Request

Ladder Logic

```

+----[ XIC B3:0/0 ]-----+
|   Auto_Run_Request   |
|-----+----[ XIC B3:0/3 ]-----+----( OTE B3:0/2 )----|
|   |               |   System_Ready      |   Pump_Run_Request
+----[ XIC B3:0/1 ]-----+
|   Manual_Run_Request

```

Rung Description:

Either Auto Run Request or Manual Run Request can enable Pump_Run_Request, provided System Ready remains true.

Engineering Purpose:

I combined both mode-specific requests into one common pump request so they share the same timer and motor-output logic.

8.10 Rung 0009 - Pump Startup Delay

Ladder Logic

```
|-----[ XIC B3:0/2 ]-----[ TON T4:0 ]-----|
      Pump_Run_Request          Delay_Timer   Time Base: 1.0 s   Preset: 3
```

Rung Description:

When Pump_Run_Request becomes true, T4:0 accumulates. After three seconds of continuous enable, T4:0/DN becomes true. Loss of the request resets the non-retentive timer.

Engineering Purpose:

The timer provides the implemented delay between acceptance of a pump request and energization of the physical pump output.

8.11 Rung 0010 - Pump Motor Output

Ladder Logic

```
|-----[ XIC T4:0/DN ]-----[ XIC B3:0/2 ]----- ( OTE O:2/0 )-----|
      Delay_Timer_DN          Pump_Run_Request          Pump_Motor
```

Rung Description:

Pump_Motor energizes when the startup delay is complete and Pump_Run_Request remains active. Loss of either condition removes the OTE output.

Engineering Purpose:

Rechecking Pump_Run_Request after the timer prevents a completed timing state from driving the motor without an active request.

8.12 Rung 0011 - Running Lamp

Ladder Logic

```
|-----[ XIC O:2/0 ]----- ( OTE O:2/1 )-----|
      Pump_Motor          Running_Lamp
```

Rung Description:

Running_Lamp directly follows the Pump_Motor command.

Engineering Purpose:

I included the lamp to provide a direct visual indication that the PLC is commanding the pump ON.

8.13 Rung 0012 - Low Level Indication

Ladder Logic

```
|-----[ XIC I:1/7 ]----- ( OTE O:2/4 )-----|
      Low_Level_Switch          Low_Level_Lamp
```

Rung Description:

Low_Level_Lamp energizes when Low_Level_Switch is true and turns OFF when the switch becomes false.

Engineering Purpose:

This provides direct operator indication of the implemented low-level switch state.

8.14 Rung 0013 - High Level Indication

Ladder Logic

```
|-----[ XIC I:1/8 ]-----|-----|-----|
           High_Level_Switch                High_Level_Lamp
```

Rung Description:

High_Level_Lamp follows the state of High_Level_Switch.

Engineering Purpose:

The dedicated lamp makes the high-level process condition directly visible to the operator.

8.15 Rung 0014 - Dry Run Monitoring Timer

Ladder Logic

```
|-----[ XIC O:2/0 ]-----[ XIO I:1/7 ]-----|-----|-----|
           Pump_Motor           Low_Level_Switch                Dry_Run_Timer   Time Base: 1.0 s   Preset: 10
```

Rung Description:

Dry_Run_Timer runs when Pump_Motor is ON while Low_Level_Switch is false. If the condition persists for ten seconds, T4:1/DN becomes true.

Engineering Purpose:

I included timer-based monitoring to detect continued pump operation without the expected Low Level Switch state.

8.16 Rung 0015 - Dry Run Fault Latch

Ladder Logic

```
|-----[ XIC T4:1/DN ]-----|-----|-----|
           Dry_Run_Timer_DN                Dry_Run_Fault
```

Rung Description:

When Dry_Run_Timer reaches Done, Dry_Run_Fault B3:1/0 is latched. The bit then participates in the master fault latch on Rung 0001.

Engineering Purpose:

I used a latched diagnostic bit so a detected dry-run event remains available to the master fault architecture.

Implementation Note:

The final screenshots do not show an OTU for B3:1/0; no reset path is added to the documentation.

8.17 Rung 0016 - Fault Lamp

Ladder Logic

```
|-----[ XIC B3:0/4 ]-----|-----|-----|
           Fault_Active                Fault_Lamp
```

Rung Description:

Fault_Lamp energizes whenever Fault_Active is true and remains ON while the master fault remains latched.

Engineering Purpose:

This provides a dedicated visual indication of the master PLC fault state.

8.18 Rung 0017 - Alarm Horn**Ladder Logic**

```
|----[ XIC B3:0/4 ]-----| ( OTE 0:2/3 )----|
      Fault_Active                Alarm_Horn
```

Rung Description:

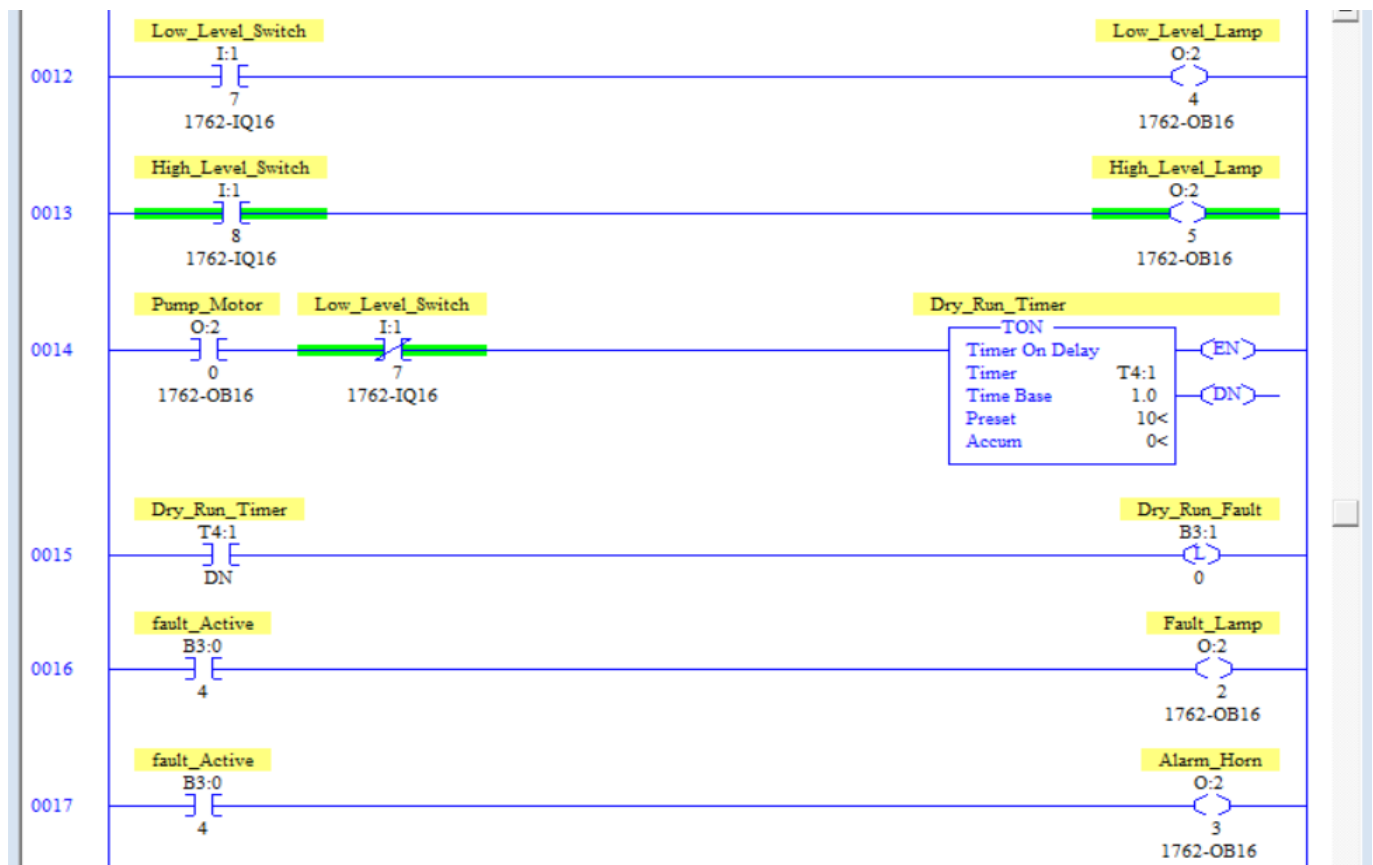
Alarm_Horn follows the master Fault_Active condition.

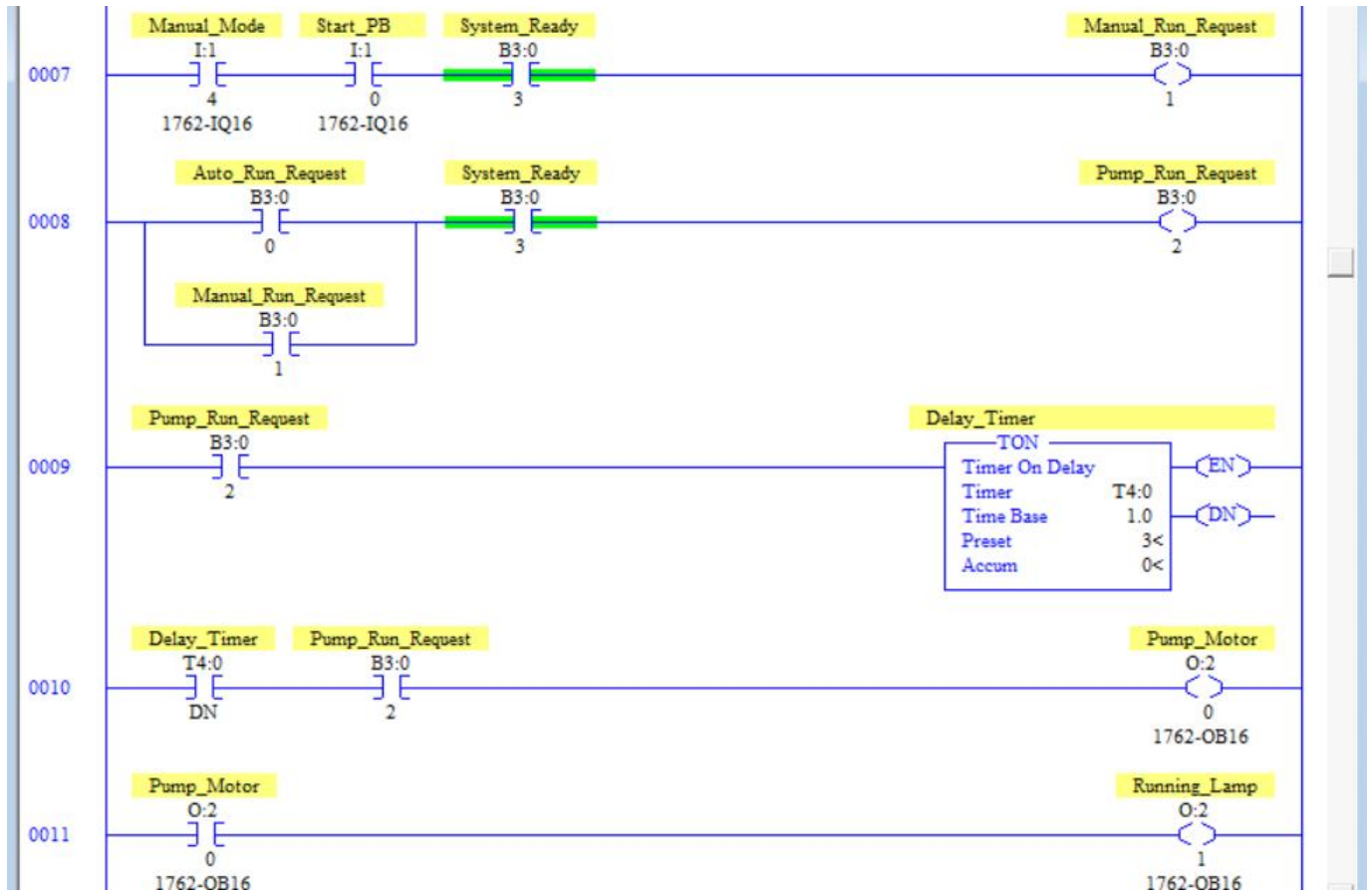
Engineering Purpose:

I added audible indication so the master fault condition can attract operator attention in addition to the visual Fault Lamp.

8.19 Actual RSLogix Implementation

The following figures show the final RSLogix implementation used as the technical source of truth for this report.





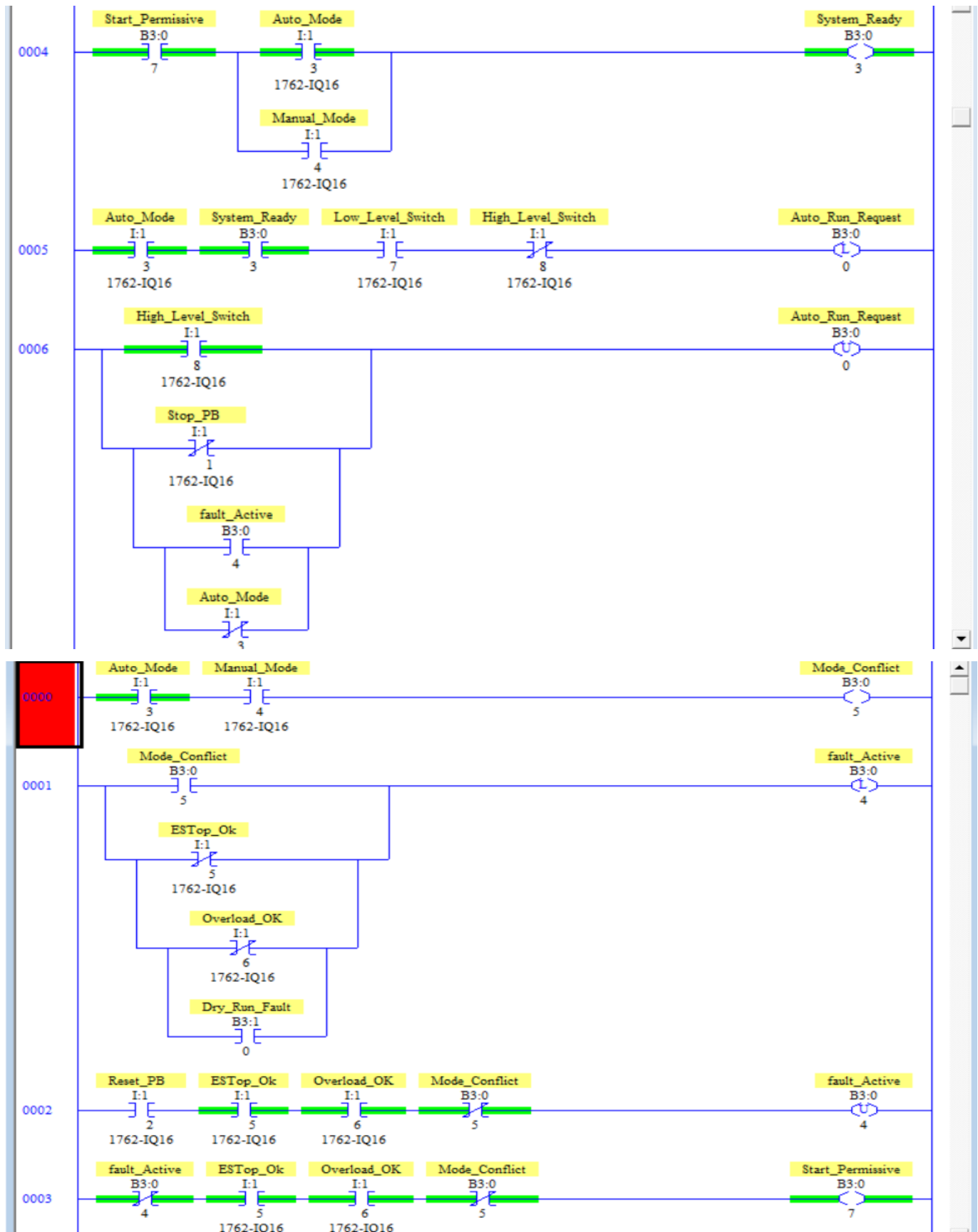


Figure P3V2.1 - Final RSLogix Implementation

9. Sequence of Operation

Step 1 - Initial Condition: The PLC continuously monitors operating-mode selection, E-stop status, overload status, and the master fault condition. Pump_Motor remains OFF until a valid Pump Run Request completes the startup delay.

Step 2 - Mode Selection: The operator selects either Auto mode or Manual mode.

Step 3 - Mode Conflict Detection: If Auto and Manual modes are selected simultaneously, Mode_Conflict becomes true.

Step 4 - Master Fault Evaluation: Mode Conflict, an unhealthy E-stop condition, an unhealthy overload condition, or Dry_Run_Fault can latch Fault_Active.

Step 5 - Start Permissive: When Fault_Active is false, EStop_OK and Overload_OK are healthy, and Mode_Conflict is false, Start_Permissive becomes true.

Step 6 - System Ready: With Start_Permissive true and either Auto or Manual selected, System_Ready becomes true.

Step 7 - Automatic Pump Request: In Auto mode, System_Ready, Low_Level_Switch true, and High_Level_Switch false allow Auto_Run_Request to latch.

Step 8 - Automatic Stop Conditions: Auto_Run_Request is unlatched when High_Level_Switch is true, the implemented Stop condition occurs, Fault_Active becomes true, or Auto mode is removed.

Step 9 - Manual Pump Request: In Manual mode, Start_PB together with System_Ready energizes Manual_Run_Request.

Step 10 - Common Pump Request: Either mode-specific request can energize Pump_Run_Request while System_Ready remains true.

Step 11 - Pump Startup Delay: Pump_Run_Request enables T4:0 for three seconds.

Step 12 - Pump Motor Start: When T4:0/DN and Pump_Run_Request are true, Pump_Motor energizes.

Step 13 - Running Indication: Running_Lamp follows Pump_Motor.

Step 14 - Level Indication: Low_Level_Lamp and High_Level_Lamp follow their corresponding level switches.

Step 15 - Dry-Run Monitoring: While Pump_Motor is ON and Low_Level_Switch is false, T4:1 accumulates.

Step 16 - Dry-Run Timeout: If the condition persists for ten seconds, T4:1/DN becomes true.

Step 17 - Dry-Run Fault: T4:1/DN latches Dry_Run_Fault.

Step 18 - Master Fault Response: Dry_Run_Fault participates in Rung 0001 and causes Fault_Active to latch.

Step 19 - Pump Shutdown: Fault_Active removes Start_Permissive, causing System_Ready and Pump_Run_Request to drop and Pump_Motor to turn OFF.

Step 20 - Fault Indication: Fault_Active energizes Fault_Lamp and Alarm_Horn.

Step 21 - Master Fault Reset: When Rung 0002 conditions are satisfied, Reset_PB can unlatch Fault_Active. The final ladder does not show a Dry_Run_Fault OTU.

10. Testing & Validation

The final Project 3 Version 2 ladder represents the tested implementation in which the project requirements passed. The verification table below is aligned to the actual final ladder rather than the unrelated Project 2 supporting documents.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Establish healthy E-stop/overload conditions and select one valid mode	Start_Permissive and System_Ready become true	Ready-state logic operated as required	PASS
T-02	Select Auto and Manual simultaneously	Mode_Conflict becomes true; Fault_Active latches	Mode conflict detected correctly	PASS
T-03	Restore valid mode and perform valid fault reset	Fault_Active unlatches when Rung 0002 conditions are true	Master fault reset operated correctly	PASS
T-04	Auto mode: Low Level true and High Level false	Auto_Run_Request latches	Automatic request established	PASS
T-05	Maintain automatic request	T4:0 accumulates for 3 s	Startup delay operated correctly	PASS
T-06	Allow T4:0/DN to become true	Pump_Motor and Running_Lamp energize	Pump started after delay	PASS
T-07	Make High Level Switch true	Auto_Run_Request unlatches and pump stops	High-level stop operated correctly	PASS
T-08	Activate implemented Stop condition	Auto_Run_Request unlatches	Stop behavior verified	PASS
T-09	Remove Auto mode during automatic request	Auto_Run_Request unlatches	Mode-removal shutdown verified	PASS
T-10	Operate in Manual mode using Start_PB	Manual_Run_Request follows enabling conditions	Non-latched Manual operation verified	PASS
T-11	Operate Low Level Switch	Low_Level_Lamp follows I:1/7	Low-level indication correct	PASS
T-12	Operate High Level Switch	High_Level_Lamp follows I:1/8	High-level indication correct	PASS
T-13	Make EStop_OK false	Fault_Active latches; pump is inhibited/stopped	E-stop monitoring response verified	PASS
T-14	Make Overload_OK false	Fault_Active latches; pump is inhibited/stopped	Overload response verified	PASS
T-15	Run pump with Low Level Switch false	T4:1 accumulates toward 10 s	Dry-run timer operated correctly	PASS
T-16	Maintain dry-run condition through timeout	Dry_Run_Fault and Fault_Active latch	Dry-run fault response operated correctly	PASS
T-17	Generate master fault	Fault_Lamp and Alarm_Horn energize	Visual and audible fault indication verified	PASS

Validation Summary

Testing confirmed the major operating and fault functions represented in the final ladder. Auto mode used the implemented Low Level and High Level conditions to establish and remove an automatic pump request, while Manual mode provided a separate non-latched operator-controlled request.

Both modes used the same common Pump Run Request and three-second Delay Timer before the physical pump output energized. E-stop, overload, and mode-conflict conditions were incorporated into the master fault architecture, while Fault Lamp and Alarm Horn followed the latched Fault Active state.

Dry-run monitoring operated according to the final ladder. Pump operation with Low Level Switch false allowed T4:1 to accumulate, and maintaining that condition through the ten-second preset caused Dry Run Fault and the master fault response to activate.

The testing represents PLC program validation in the project development/simulation environment. It does not represent commissioning of a physical pumping system or validation of a safety-rated machine-control system.

11. Results

Project 3 Version 2 successfully expanded the original water-tank control concept into a structured pump-control application.

The final program supports Auto and Manual operating modes, detects simultaneous mode selection, establishes Start Permissive and System Ready conditions, separates operating requests from the physical pump output, and applies a three-second startup delay before pump energization.

Automatic operation uses the implemented low- and high-level switch conditions. A valid low-level condition can establish the automatic pump request, while High Level Switch provides an automatic stop condition through the Auto Run Request reset logic. Manual operation uses a separate non-latched request but shares the common Pump Run Request, Delay Timer, Pump Motor output, and fault architecture.

The system provides direct Low Level and High Level indication through separate output lamps. The dry-run monitoring function adds diagnostic behavior: if the pump remains commanded ON while Low Level Switch is false for ten seconds, Dry Run Fault latches and causes the master Fault Active condition to latch.

Fault Active removes operating permission and provides visual and audible fault indication through Fault Lamp and Alarm Horn. All defined functional requirements represented by the final tested ladder passed in the PLC development environment.

12. Challenges & Troubleshooting

A useful troubleshooting path for Project 3 Version 2 is: Mode / Protective Inputs -> Mode_Conflict / Fault_Active -> Start_Permissive -> System_Ready -> Auto/Manual Run Request -> Pump_Run_Request -> T4:0 -> Pump_Motor.

For level and dry-run problems, continue through: Low/High Level Inputs -> Auto_Run_Request -> Pump_Motor -> Low_Level_Switch -> T4:1 -> Dry_Run_Fault -> Fault_Active.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
System Ready does not turn ON	Fault active, E-stop/overload unhealthy, or no mode selected	Monitor B3:0/4, I:1/5, I:1/6, I:1/3 and I:1/4	Restore healthy conditions and select one valid mode
Mode Conflict fault occurs	Auto and Manual selected together	Monitor I:1/3, I:1/4 and B3:0/5	Select only one mode and perform valid reset
Auto request does not latch	Auto mode, System Ready, Low Level, or High Level condition incorrect	Trace Rung 0005	Verify all enabling contacts
Auto pump stops unexpectedly	High Level, Stop, Fault Active, or Auto mode removal	Trace Rung 0006	Identify which reset branch became true
Manual request does not energize	Manual mode, Start_PB, or System Ready missing	Monitor Rung 0007	Verify all three enabling conditions
Delay Timer does not accumulate	Pump_Run_Request is false	Monitor B3:0/2 and T4:0/EN	Trace Auto/Manual request and System Ready
Timer done but pump remains OFF	Pump_Run_Request was removed	Monitor T4:0/DN and B3:0/2	Restore a valid operating request

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Low/High Level lamp incorrect	Corresponding level-switch state incorrect	Monitor I:1/7 or I:1/8 and outputs	Verify simulated/field level input
Dry Run Timer does not accumulate	Pump Motor OFF or Low Level Switch true	Monitor O:2/0, I:1/7 and T4:1/EN	Verify Rung 0014 conditions
Dry Run Fault activates	Pump remains ON while Low Level Switch is false for 10 s	Monitor T4:1.ACC, T4:1/DN and B3:1/0	Investigate process condition or level-switch signal
Fault Lamp/Horn remain ON	Fault_Active remains latched	Monitor B3:0/4	Identify active fault source and perform valid master reset

A key troubleshooting lesson is to trace the pump command from the upstream conditions rather than beginning at the output. A pump that remains OFF may have no output problem; the cause can originate in mode selection, permissive logic, level demand, Manual request, or startup timing.

13. Lessons Learned

This project strengthened my understanding of how a process condition such as tank level can be converted into a structured equipment request rather than controlling a physical output directly.

I learned how a latched Auto Run Request can preserve process demand after the initiating level condition, while a dedicated unlatch rung defines the conditions that terminate the automatic cycle.

The project also demonstrated the difference between Auto and Manual command behavior. Automatic operation uses a maintained internal request, while Manual operation uses an OTE and therefore requires its command conditions to remain true.

Using a common Pump Run Request showed how mode-specific logic can be separated from shared equipment-control functions. Both Auto and Manual modes use the same startup timer and Pump Motor output logic.

The level-indication rungs reinforced the value of exposing process state independently from equipment commands. Dry-run monitoring introduced a process diagnostic based on the relationship between commanded pump operation and an expected level condition.

The project also reinforced the importance of documenting exactly what the final PLC file implements. The visible final ladder contains a latched Dry Run Fault but does not show a corresponding OTU instruction, so the report preserves that behavior rather than silently correcting it.

14. Industrial Relevance

Automatic tank and pump control is used in water systems, process industries, chemical handling, food and beverage production, wastewater applications, utility systems, and many other industrial processes.

This project demonstrates controls concepts relevant to those applications: level-based automatic demand, Manual operation, mode management, centralized permissives, delayed equipment startup, process indication, and dry-run monitoring.

Separating the level condition from the physical Pump Motor command is important from a controls-architecture perspective. The process determines whether pumping is requested, while the PLC separately determines whether the request is permitted and when the physical output can energize.

Dry-run monitoring is also relevant because pumps can be damaged or operate improperly when expected process conditions are absent. The implemented timer-based diagnostic demonstrates the basic concept of detecting that abnormal condition.

The project remains a PLC training/simulation implementation. A real pump installation would normally require additional engineering such as properly designed motor protection, pump feedback, pressure or flow instrumentation where required, safety-rated systems where applicable, field-device failure diagnostics, and commissioning using the actual process equipment.

15. Version 1 -> Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Benefit
Basic tank-level control	Auto and Manual operating modes	Provides distinct process and operator-control behaviors
Directer level-to-output logic	Level condition -> Auto Run Request -> Pump Run Request -> timer -> Pump Motor	Separates process demand from equipment command
Limited operating checks	Start Permissive and System Ready	Improves operating-state control and troubleshooting
No mode-conflict handling	Mode_Conflict detection	Identifies invalid simultaneous mode selection
Limited fault architecture	Latched Fault_Active	Provides one master abnormal state
Basic automatic control	Latched Auto Run Request with dedicated stop conditions	Creates a defined automatic operating cycle
Limited manual control	Non-latched Manual Run Request	Provides separate operator-controlled operation
Immediate equipment command	Three-second Delay Timer	Adds controlled startup timing
Basic level sensing	Dedicated Low and High Level lamps	Improves process-state visibility
No dry-run diagnostic	T4:1 Dry Run Timer	Detects prolonged pump operation without expected level state
No persistent dry-run indication	Latched Dry_Run_Fault	Preserves abnormal process event
Limited fault indication	Fault Lamp and Alarm Horn	Provides visible and audible abnormal-state indication

The main engineering progression is the separation of process demand, operating permission, equipment request, timing, physical output, and diagnostics. Version 1 established the foundation of automatic water-level control; Version 2 reorganizes that requirement into a more maintainable PLC architecture with operating modes, fault handling, process indication, and dry-run monitoring.

16. Project Summary

Project 3 Version 2 upgraded the original water-tank control project into a structured Water Tank Pump Control System using an Allen-Bradley MicroLogix 1100 PLC and RSLogix Micro Starter Lite.

The final ladder includes Auto and Manual operating modes, Mode Conflict detection, master fault latching, conditional fault reset, Start Permissive, System Ready status, automatic level-based pump requests, Manual pump operation, common Pump Run Request logic, a three-second startup delay, Pump Motor control, running indication, low- and high-level indication, dry-run monitoring, Fault Lamp, and Alarm Horn.

In Auto mode, the Low Level and High Level switches determine when the automatic pump request is established and removed. Manual mode provides a separate non-latched command while sharing the same downstream pump-control sequence.

The pump starts only after the common Pump Run Request remains active through the configured three-second delay. While the pump is operating, the PLC monitors Low Level Switch status through the ten-second Dry Run Timer. A sustained dry-run condition latches Dry Run Fault and causes the master Fault Active response.

The completed project demonstrates my progression from basic tank-level control toward a more structured PLC process-control architecture using operating modes, permissives, internal requests, delayed output control, process-state indication, diagnostics, and master fault handling.

The project was developed and tested in the PLC simulation/development environment. The final documentation preserves the exact implemented logic shown in the frozen ladder, including the visible latched Dry Run Fault behavior.